

Physics Concepts as Indicative Framework to Model Systems of Economies Ex-ante

Coincise Notes and Conjectures Explorations of Natural Laws as Approach to Preliminarily Structure Economies

Eugenio Net

2026-06-26

Distillation and exploratory attempt ... Interconnection network between movement, force, momentum, energy, and work. Describing systems with Lagrangian and Hamiltonian when constraints and starting conditions exist. Modeling systems and optimisation with constraints and conditions. Interwoven system of macro- and microeconomy under constraints, limitations, and conditions.

Table of contents

Potentials	3
Potential as “intrinsic energy of inertia”	3
Electro-Magnetic Field (force caused by potential)	3
Electro-Magnetic interactions/coupling (Maxwell conditions)	5
Potential derivation	5
elec. Potential structure	6
magn. Potential structure	6
Electric endogen energy	6
Electric induced energy (exogenous excitation)	6
Power:	7
Frequency:	7
Phase:	7
Phaseshift:	7
Refraction:	7
Dispersion:	8
Dielectricity:	8
Permeability:	8
Conductivity:	8

Conductor (charge conductive element for energy transfer) condition:	8
Slow changing current (quasi-stationary, low frequencies) condition:	8
Energy Balance	8
Local Force Balance	8
Local Momentum Conservation	9
Tension Tensor	9
Mechanical	9
Electromagnetic energy transfer	10
Thermal	11
Gravitational	11
Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potential difference, tension):	11
Low Frequency	11
High Frequency	12
Continuity at Rest: Electric Charge	12
Continuity at Movement: Magnetic Field	12
Continuity at Acceleration: Radiation Energy	12
Continuity and periodic (harmonious oscillation) Movement in VACUUM	12
Continuity and periodic (harmonious oscillation) Movement in MATTER	12
Near Field	13
Far Field	13
Dissipation in Vacuum	13
Dissipation in Matter	13
Energy gain by excitation	13
Frequency Cause-Effect	15
Frequency (commonness, probability, likelihood)	15
Frequency contextual Causes	15
Conditions (skin, interface, border, surface)	18
Boundary Conditions	18
Interior Conditions	18
Abbreviations	18
Distinguishing Situations Conditions and Dependencies	18
Spacetime Embedding	18
Movement Framework	19
Medium Property	19
Field Distance	19
Localisation Identity	19
Operators	20
Acronyms	20
Acronyms of Physics: ?@sec-workenergy	20
Acronyms of Economy: ?@sec-productivityvalue	21

Potentials

Potential as “intrinsic energy of inertia”

Electro-Magnetic Field (force caused by potential)

- static: localized source charge at rest, only spacial dependency, no time dependency
 - stationary: source charge at uniform movement, steady current flow, space and time dependency
 - dynamic: source charge at fast movement, radiation, space and time dependency
 - relativistic: source charge at very speedy movement, radiation, space and time dependency
-
- rest localization vs. time-dependency
 - slow vs. fast movement
 - vacuum vs. matter
 - local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
 - macro vs. micro
 - far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := \text{rot}(\mathcal{A}) \stackrel{\text{Matter}}{=} \mu_0 (\mathcal{H} + \mathcal{M}) \stackrel{\text{Vacuum}}{=} \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left(c p_{0_{dipol}} + m_0 \right) \quad (1)$$

Magnetic Field in Matter

Excitation $\hat{=}$ Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservativ}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} - \frac{v}{c}\right)\right]}{r^3}}_{\text{acceleration}} \right] \quad (3)$$

Electric Field in Matter

Displacement $\hat{=}$ Excitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Excitation:} \quad \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \stackrel{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \langle p_m \rangle \quad (4)$$

Abbreviations

Φ = electric potential A = magnetic potential $j = \nabla \times \mathcal{H}$ = elec. current density

$\mathcal{E}$ = Electric Field $\mathcal{B}$ = Magnetic Field $\nabla \cdot \mathcal{B} = 0$ $\mathcal{D}$ = Displacement density of dielectricum $\mathcal{M}$ =

$\mathcal{P}$ = Polarisation $\mathcal{H}$ = magn. excitation, magn. field strength ("magn. displacement")

n = spacial molecule density in volume

$$\alpha = 4\pi r^3 = \text{constant} \quad \chi = \text{elec. susceptibility of dielectricum} \quad \kappa = \frac{\mu - \mu_0}{\mu_0} = \text{magn. susceptibility}$$

$$\langle p_m \rangle = \text{spacial average of magn. dipol moment} \quad \langle \mathcal{E}_m \rangle = \text{spacial average of electric field}$$

$$\bar{\mathcal{E}}_m = \text{time average of electric field} \quad \mu = \text{magn. permeability} \quad \mu_0 = \text{magn. constant ...}$$

$$\varepsilon = \text{dielectricity constant} \quad \varepsilon_0 = \dots$$

Electro-Magnetic interactions/coupling (Maxwell conditions)

conservativ, static charge localisation:

...

stationary, slow charge movement:

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \varepsilon_0 \mathcal{E} \quad (5)$$

relativistic, fast charge movement:

...

Potential derivation

⇒ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} U = J \quad (6)$$

⇒ Potential caused by “ignition J” (inhomgenity) from source at rest, $v = 0$:

$$U = -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \quad (7)$$

elec. Potential structure

$$J = -\frac{1}{\varepsilon_0} \varrho$$

$$U_e := \Phi \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\varepsilon_0} \int \frac{\varrho}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} It \quad (8)$$

magn. Potential structure

$$J = -\mu_0 j$$

$$U_m := A \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2 \varepsilon_0} I = \frac{1}{c^2} \frac{1}{t} U_e \stackrel{\text{source in uniformal movement, } v=\text{const}}{=} \quad (9)$$

Electric endogen energy

- Bond Energy
- Binding Energy

$$E_{pot} = mc^2$$

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (10)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (11)$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Excitation Energy

$$E_{kin} = \frac{1}{2} mv^2 \hat{=} (pc)^2 \hat{=} h\nu = \hbar\omega$$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2 \quad (12)$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (13)$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2} C U^2 \quad (14)$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2} L I^2 \quad (15)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (16)$$

Frequency:

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda \frac{1}{T} = \sqrt{\frac{k}{m}} \stackrel{vacuum}{=} kc \stackrel{matter}{=} k \frac{c}{\eta} = kc \frac{\tan \beta}{\tan \alpha} \stackrel{electromagnetic}{=} kc \left(\mathcal{E}_\beta^{-1} \mathcal{E}_\alpha \right)^{-1} = kc \frac{\varepsilon_\beta}{\varepsilon_\alpha} = kc \sqrt{\frac{\varepsilon_0 \mu_0}{\varepsilon \mu}} \quad (17)$$

Phase:

$$\varphi = k \left(\vec{r} \cdot \vec{k} \right) - \omega t := ks - \omega t = \dots \quad (18)$$

Phaseshift:

$$\Delta\varphi = \Delta k \left(\vec{r} \cdot \vec{\Delta k} \right) - \Delta\omega t := \Delta ks - \Delta\omega t = \dots \quad (19)$$

Refraction:

Refraction $\Leftrightarrow \exists$ Matter; refraction causes dispersion

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\lambda_0}{\lambda} = c \sqrt{\varepsilon \mu} \hat{=} \mathcal{E}_\beta^{-1} \mathcal{E}_\alpha = \frac{\varepsilon_\alpha}{\varepsilon_\beta} = \dots \quad (20)$$

Dispersion:

Dispersion $\Leftrightarrow \exists$ Matter (dielectric medium or conductor); Dispersion $\neq$ Dissipation $\neq$ Diffusion

$$\dots \quad (21)$$

Dielectricity:

$$\varepsilon(\omega) = \dots \quad (22)$$

Permeability:

$$\mu(\omega) = \dots \quad (23)$$

Conductivity:

$$\sigma(\omega) = \dots \quad (24)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \quad (25)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (26)$$

Energy Balance

$$\frac{\partial}{\partial t} \sum w_i + \cdot \sum \mathcal{S}_{w_i} + j \cdot \mathcal{E} = 0 \quad (27)$$

Local Force Balance

$$\dot{p} = F \hat{=} \underbrace{\rho \mathcal{E} + j \times \mathcal{B}}_{\text{charge force}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force}} + \underbrace{\nabla \cdot \mathcal{T}}_{\text{tension force}} = 0 \quad (28)$$

Local Momentum Conservation

linear momentum conservation if $p = \text{const.}$, otherwise “tension” occurs:

$$p = \int F = \int \frac{d}{dt} p \Rightarrow F = \dot{p} \hat{=} \int_{\mathbb{R}^3} \frac{d}{dt} \left[(\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S} \right] = \frac{d}{dt} \underbrace{\left[\int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{1}{c^2} \mathcal{S} \right]}_{= p \text{ momentum conserved if } p = \text{const.}} = \begin{cases} 0 \Leftrightarrow p = \\ - \int_{\mathbb{R}^2} \mathcal{T} d \end{cases} \quad (29)$$

$$\text{Momentum } p = \int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{1}{c^2} \mathcal{S}$$

$$\text{photon Spin} = \int_{\mathbb{R}^3} \frac{1}{c^2} \mathcal{S}$$

angular momentum (photon Spin):

$$D = \dot{L} \hat{=} \dots \quad (30)$$

Tension Tensor

$$\nabla \cdot \mathcal{T} = \varepsilon_0 (\mathcal{E} \times \nabla \times \mathcal{E} - \mathcal{E} \nabla \cdot \mathcal{E}) + \frac{1}{\mu_0} (\mathcal{B} \nabla \times \mathcal{B} - \mathcal{B} \nabla \cdot \mathcal{B}) \quad (31)$$

Abbreviations:

w_i = energy densities (electric w_e , magnetic w_m , gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poynting-Vectors $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{T}$ = Tension Tensor (momentum flow density); Tension $\hat{=}$ Forces = Pressure - Traction $\hat{=}$ Torsion Tension at angular momentum

Mechanical

...

Electromagnetic energy transfer

charge conservation:

$$\frac{\partial}{\partial t} \varrho + \nabla \cdot j = 0 \quad (32)$$

$\varrho = \sigma \delta$ = charge density in interior

σ = charge density at border (skin)

δ = Dirac delta-function

$j = \sigma (\mathcal{E} + v \times \mathcal{B}) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t} \mathcal{D} \hat{=}$ electric current density

$j \cdot \mathcal{E} \hat{=}$ energy gain by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=}$ energy transport density (Intensity = $|\bar{\mathcal{S}}|$)

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (w_e + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (33)$$

Abbreviations:

$$\Delta \cdot := \text{div grad} = \text{Laplacian}$$

J = "ignition of inhomogeneity, cause" j = Current density ϱ = Charge density

V = Potential difference I = Current R = Resistance

L = Inductance C = Capacitance Q = charge (ion) $\Psi = LI$ = mag. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H}$ = Poynting-Vector, radiation intensity

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr$ = dissipation energy lost from radiation ≈ 0 for low frequencies ω

$$d = \sqrt{\frac{1}{\mu\sigma\omega}} = \text{skin depth (surface) of conductor} \quad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"}$$

$$\mu = \text{permeability of conductor} \quad \sigma = \text{conductivity of conductor (dielectricum)}$$

Thermal

...

Gravitational

...

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potential difference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (34)$$

Low Frequency

$$\omega \ll \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \quad (35)$$

$$L \approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0$$

High Frequency

$$\omega \gg \frac{1}{\mu\sigma} \frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \quad (36)$$
$$L(t) \neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Continuity at Rest: Electric Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emergence of Radiation

Continuity and periodic (harmonious oscillation) Movement in VACUUM

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \nu \lambda = \frac{\lambda}{T}$$
$$2\pi = k\lambda = T\omega = \frac{\omega}{\nu}$$

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \quad \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

Continuity and periodic (harmonious oscillation) Movement in MATTER

$$\text{Refraction index } \eta \text{ emerges: } \varepsilon\mu = \left(\frac{\eta}{v_{ph}}\right)^2 \hat{=} \left(\frac{\eta}{c}\right)^2$$

$$\text{Dispersion: } k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$$

Near Field

static, vacuum, granular spacetime

$$r \ll \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\epsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$\vec{p}$ = elec. Dipol

Far Field

static, vacuum, flat spacetime

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\epsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|} \right)^2}{\frac{\epsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|} \right)^2} = 1$$

Dissipation in Vacuum

no Continuity of charge/matter, no Refraction index, Reflection

Dissipation in Matter

no Continuity of charge/matter with dielectricum, refraction merges, reflection, absorption

Energy gain by excitation

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

¹Wirkungsquantum

$$h = \tan(\varphi) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \quad (37)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (38)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (39)$$

V_0 = Opposing Potential ($I = 0$)

$\varphi = \angle(E_{kin}, \nu)$ e = electron charge Q = ion charge

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{C}{L} R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$$

Ionisation Energy of chemical element $E_i = h\nu = h \frac{\lambda}{c} = \hbar\omega = \frac{\hbar}{k\lambda}\omega$

Energy released (gained), see... Equation 6

- with mass (e.g. electron), Tunnel-Effect: $E_{gain} = E_{pot} - E_i = mc^2 - h\nu_i$
- massless (e.g. photon), Doppler-Effect (frequency shift): $E_{gain} = E_{kin} - E_i = h(\nu_0 - \nu_i) = \Delta\omega\hbar$

Frequency Cause-Effect

Frequency (commonness, probability, likelihood)

$$\omega = \frac{d}{dt}\varphi = \eta kv = kc = \frac{2\pi}{\mathcal{T}} = 2\pi \cdot \nu = \frac{a}{v} = \frac{v}{r} = \frac{E}{\hbar} = \sqrt{\frac{k}{m}} = \sqrt{\frac{r \times F}{m}} = \sqrt{\frac{g}{l}} \stackrel{\text{electromagnetic}}{=} \dots = \sqrt{\frac{1}{LC}} \stackrel{QM}{=} \frac{1}{2} \frac{p^2}{m \hbar} \quad (40)$$

$$\nu = \frac{E}{h} = \frac{v_{phs}}{\lambda} = \frac{c^2}{v} \frac{1}{\lambda} = \frac{\omega}{k} \frac{1}{\lambda} = \frac{\omega}{2\pi} \quad (41)$$

$$\nu_i = \frac{eV_{0i}}{h} + \nu_0 = \frac{E_{kin}}{h} \omega \mathcal{T} + \nu_0 \quad (42)$$

$$e = \text{electron charge} \quad V_0 = \text{Opposing Potential } (I = 0) \quad (43)$$

Energy Level: $i = [1, n] \in \mathbb{N}$

$$E_{kin}^{max} = eV_0 = h(\nu - \nu_0) \quad (44)$$

Frequency contextual Causes

Frequency of rotational velocity as periodic occurrence of an event/state by space distance and time periodicity as well as movement speed and reach length, $v^2 = v_{ph} v_{gr}$, $\omega t = 2\pi \nu t = \lambda \neq 0$:

$$\frac{v}{r} = \omega = \frac{\lambda}{t} \quad (45)$$

Rest state of source charge

Statics localization of charge. Only spacial demependency, no time dependency.

Energy transport ... Information transmission ...

$$v = 0 \quad \dot{v} = 0$$

Slow Movement of source charge

Stationarity localization of current flow by slow movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v = \text{const.} \ll c \quad \dot{v} = 0$$

Fast Movement of source charge

Fields forces by fast movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \approx c \quad \dot{v} = 0$$

Acceleration Movement of source charge

Radiation energy by accelerated movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \neq 0 \quad \dot{v} \neq 0$$

Movement in Vacuum

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{ph}} = 1 \quad 2\pi = k\lambda \quad v_{ph} = c \quad \omega = kc$$

$$\varepsilon_0 \mu_0 = \frac{1}{c^2}$$

$$\text{Continuity: } \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

Movement in Matter

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{ph}} = \sqrt{\frac{\varepsilon \mu}{\varepsilon_0 \mu_0}} \neq 1 \quad v_{ph} = \pm \sqrt{\frac{1}{\varepsilon \mu} - \frac{\sigma^2}{4\varepsilon^2 k^2}} \quad \omega = kv_{ph} = kc \sqrt{\frac{\varepsilon_0 \mu_0}{\varepsilon \mu}} =$$

$$kc \frac{\tan \beta}{\tan \alpha} = k \frac{c}{\eta}$$

$$\varepsilon \mu = \left(\frac{\eta}{c}\right)^2$$

$$\text{Dispersion: } k^2 = \varepsilon \mu \omega^2 + i\mu \sigma \omega$$

Near Field

$$r_0 \ll r \ll \lambda \quad kr \ll 2\pi \quad \frac{w_m}{w_e} \ll 1$$

r is small, fast movement

Far Field

$$r \gg \lambda \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1 \quad \frac{w_m}{w_e} = 1$$

r is big, slow movement

High Frequency

$$\omega \gg \frac{2}{\mu\sigma r_0^2}$$

λ is big, ν frequency is small, short time period

Low Frequency

$$\omega \ll \frac{2}{\mu\sigma r_0^2}$$

λ is small, ν frequency is big, long time period

Conditions

Border (skin surface): ...

Interior (medium body): ...

Global:

Local:

Consequences

dielectricity constant (matter medium): $\varepsilon = \varepsilon(\omega) = \mathcal{E}^{-1} \mathcal{D} = \mathcal{E}^{-1} (\mathcal{P} + \varepsilon_0 \mathcal{B})$

permeability constant (matter medium): $\mu = \mu(\omega) = \mathcal{B}^{-1} \mathcal{H} = \mathcal{B}^{-1} \left(\frac{1}{\mu_0} \mathcal{B} - \mathcal{M} \right)$

conductivity (matter medium): $\sigma = \sigma(\omega) = \frac{\sigma_0}{\sqrt{1 + \omega^2 \tau^2}}$

magn. energy density (vacuum): $w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2$

elec. energy density (vacuum): $w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2$

wave vector: $k = \frac{2\pi}{\lambda} = \eta \frac{\omega}{c}$

refraction index (matter medium): $\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\eta_{medium1}}{\eta_{medium2}}$

Energy: $E = h\nu = \hbar\omega = E_{kin} + E_{pot} \stackrel{!}{=} \sqrt{(pc)^2 + (mc^2)^2}$

Conditions (skin, interface, border, surface)

Boundary Conditions

$$\mathbf{n} \cdot [\mathcal{D}] = \sigma$$

$$\mathbf{n} \cdot [\mathcal{B}] = 0$$

$$\mathbf{n} \times [\mathcal{E}] = 0$$

$$\mathbf{n} \times [\mathcal{H}] = J$$

Interior Conditions

e.g. for Superconductivity

$$\mathbf{n} \cdot \mathcal{H} = 0$$

$$\mathbf{n} \times \mathcal{D} = 0$$

Abbreviations

$\mathbf{n}$ = normal vector (orthogonality orientation)

$\mathcal{E}$ = elec. Field (“traction force”?)

$\mathcal{B}$ = magn. Field (“compressive force”?)

$\mathcal{D}$ = elec. Displacement Field (“exitioan”)

$\mathcal{H}$ = magn. Excitation Field (“incentivation”)

σ = electric charge at surface

J = current density at surface

Distinguishing Situations Conditions and Dependencies

effects of charge (separaton) and differentiating situations (conditioning frameworks) for fields and potentials:

Spacetime Embedding

- Spacial dependence (mechanics)
- Time dependence (dynamics)

Movement Framework

- Static: rest system, equilibrium
- Dynamic:
 - Stationary: slow motion
 - Relativistic: fast motion
 - Radiation; accelerated motion

Medium Property

- Vacuum: spacial mean = time mean
- Matter:
 - Mass (inertia)
 - Charge (conductor, ion)
 - Dielectricum (isolator)
 - Dispersion (refraction)

Field Distance

- Near Field (granular spacetime, mean average values)
- Far Field (flat spacetime)

Localisation Identity

- Global (regular)
- Local (singular)

-> Tool concepts: Momentum, Force, Energy

-> Conditions framework: - continuity condition (charge, current) - equilibrium conditions (action vs. reactio, homogeneity vs. inhomogeneity) - initial conditions (starting point) - border (skin) conditions (differentiation) - interior conditions (integration) - conservations of local densities (charge, energy, momentum)

-> Properties: - characteristics from charge: dielectricity ε , permeability μ , conductivity σ , frequency ω - characteristics from heat: ... - Properties from gravitation: ...

Operators

$\nabla := grad$ = Gradient (Gefälle, Neigung)

$\nabla \cdot := div$ = Divergence (Quelle)

$\nabla \times := rot$ = Rotation (Wirbelung, Curl)

∇^2 = Lagrange Operator (Krümmung, Beschleunigung)

$\Delta \cdot := div(grad)$ = Laplace Operator

$\Delta :=$ Difference, shift

Acronyms

Acronyms of Physics: ?@sec-workenergy

A = Action

t = time

v = velocity

p = momentum

I = inertia

$\mathcal{H}$ = Hamiltonian

$\mathcal{U}$ = effective potential Energy ($\in E_{pot}$)

$\mathcal{Z} = \frac{1}{2} \frac{L_0^2}{mr^2}$ = Centrifugal Potential

k = Wave Vector ("curvature")

P = Pressure

T = endogen Temperature

U = endogen Energy ($E_{kin} + E_{pot}$)

$\mathcal{A}$ = Magnetic Potential

g_0 = Gas constant

q = charge

ϵ_0 = electric constant

μ_0 = magnetic constant

$\mathcal{E}$ = Electric Field

$c_0 = \frac{1}{m} \frac{\Delta H}{\Delta T}$ = specific heat

$E = E_{kin} + E_{pot}$ = total energy

r = dpace (distance between two points, one-dimensional length)

c = light speed

F = force

$\mathcal{L}$ = Lagrangian

$\mathcal{K}$ = kinetic Energy

$\mathcal{V}$ = potential Energy ($\in E_{pot}$)

$V = r_1 \cdot r_2 \cdot r_3$ = Volume

W = Work (done vs. received)

L_0 = angular momentum

H = exogen Heat

$\Phi = \frac{\mathcal{V}}{q}$ = Electric Potential

b_0 = Boltzmann constant

m = mass

$\$n$ = amount of objectes (particles density

$n = \frac{N}{V}$), $n = 2$ and $f = 1$

$f = 3n \pm z$ = degrees of freedom, $f = 1$ and $n = 2$

$\mathcal{B}$ = Magnetic Field

$\mathcal{HC} = m \cdot c_0 = \frac{\Delta H}{\Delta T}$ = heat capacity

$S = b_0 \cdot \ln(\Omega)$ = Entropy (macro state)

l = Moll quantity	Ω = micro states
z = amount of constraints (boundry conditions)	b_0 = Boltzmann constant
$\kappa = \frac{c_P}{c_V} = \frac{f+2}{f}$ = adiabaty	$\iota = \kappa$ adiabaty
$\iota = 0$ isobar	$\iota = 1$ isotherm
$\iota = \infty$ isochor	...
...	...

Acronyms of Economy: ?@sec-productivityvalue

T = Taxes	M = Import of Goods and Services from foreing symstes
G = Government Expenses, incl. Social Insurances	X = Export of Goods and Services to foreign system
Y = Income of Economy from Turnover	G_A = Governmental Subsidies
D_A = Depreciations (Reinvestments) on Assets	V_N = Net Naötional Production, Society NNP
N = Monetary Quantity	Q = Monetary Turnover Velocity
V_I = Gross Domestic Product $GDP = \frac{Output}{Input}$, Tradevolume	P = Price niveau (Inflation adjusted Value)
L = Wages from Labor Work (Salaries, ...)	R = Returns, Earnings, Gains
Y_A = Income of priv. Business Households (Companies, Services, Real Estate Rentals, Retained Profits)	Y_H = Income from priv. Capital Households (Interests, Coupons, Dividends, ... of priv. Assets, Investmens, Credits, Debits, Bonds, Equity)
T_A = Tax on Capital of Corprate Compaies (Business Assets)	Y_G = Governmental Income from Assets, Services, Social Institutions/Insurances
Z_G = Interests on Governmental Debt	V_S = Gross National Produkt, Society GNP
I = Investments on Assets, incl. Storage Change	R_M = Capital Earnings and Wages from Abroad (from Foreign System)
R_X Capital Earnings and Wages to Abroad (to Foreign System)	W = Expensens, costs from human and machinary work efforts

see also [1]

1. Wachter A, Hoeber H (2005) [Compendium of Theoretical Physics](#). Springer
2. Cockshott WP (2011) [Classical Econophysics](#). Routledge Advances in Experimental; Computable Economics
3. Wladyslaw W (2013) [Macroeconometric Models](#), 1st ed. Springer
4. Lichtenegger K (2015) [Schlüsselkonzepte zur Physik](#), 1st ed. Springer Spektrum
5. Padmanabhan T (2015) [Sleeping Beauties in Theoretical Physics](#), 1st ed. Springer
6. Carter M (2001) [Foundations of Mathematical Economics](#). The MIT Press
7. Michaelides PG (2025) [21 Equations that Shaped the World Economy](#), 1st ed. Palgrave Macmillan Cham
8. Pires da Cruz J (2025) [Introduction to the Physics of the Economy and Finance](#), 1st ed. Springer Cham
9. Cass D, Shell K (2014) [The Hamiltonian Approach to Dynamic Economics](#), 1st ed. Elsevier, Academic Press
10. Haven E, Khrennikov AI, Robinson T (2017) [Quantum Methods in Social Science - A First Course](#). World Scientific, Publishing Europe Ltd